

SecurePulse Mobile — Master Product & Engineering Roadmap 🚀

This document outlines the authoritative product roadmap, engineering milestones, architectural phases, and delivery commitments for **SecurePulse Mobile** (Enterprise Cybersecurity & SOC Mobile Operations Platform).

📊 High-Level Product Architecture & Milestone Timeline

```
graph TD
  subgraph "Phase 1 - 5: Foundation & Security Core"
    P1["1. Vision & Requirements<br>✅ COMPLETED"] --> P2["2. Architecture Design<br>✅"]
    P2 --> P3["3. FastAPI Backend<br>✅ COMPLETED"]
    P3 --> P4["4. Database Persistence<br>✅ COMPLETED"]
    P4 --> P5["5. Biometric & JWT Auth<br>✅ COMPLETED"]
  end

  subgraph "Phase 6 - 8: Toolchain Integrations"
    P5 --> P6["6. GitHub Integration<br>✅ COMPLETED"]
    P6 --> P7["7. Semgrep SAST Engine<br>✅ COMPLETED"]
    P7 --> P8["8. Wazuh SIEM Connector<br>✅ COMPLETED"]
  end

  subgraph "Phase 9 - 12: Mobile & Real-Time Engine"
    P8 --> P9["9. Flutter Mobile App<br>✅ COMPLETED"]
    P9 --> P10["10. REST API Client<br>✅ COMPLETED"]
    P10 --> P11["11. WebSocket Engine<br>✅ COMPLETED"]
    P11 --> P12["12. AI Security Copilot<br>✅ COMPLETED"]
  end

  subgraph "Phase 13 - 17: Production Release & Evolution"
    P12 --> P13["13. Security Hardening<br>✅ COMPLETED"]
    P13 --> P14["14. Cloud Deployment<br>✅ COMPLETED"]
    P14 --> P15["15. Automated Testing<br>✅ COMPLETED"]
    P15 --> P16["16. Play Store Release<br>✅ COMPLETED"]
    P16 --> P17["17. Post-Launch Evolution & Offline Queue<br>✅ COMPLETED"]
  end
```



Master System Architectural Topology

flowchart TB

S["SECUREPULSE"]

S --> M[" MOBILE CLIENT"]
S --> B[" CLOUD BACKEND"]
S --> SS[" SECURITY SOURCES"]
S --> P[" PROCESSING PIPELINE"]
S --> AI[" ARTIFICIAL INTELLIGENCE"]
S --> I[" INFRASTRUCTURE"]
S --> SEC[" SECURITY"]

M --> D["Dashboard"]
M --> R["Repositories"]
M --> A["Alerts"]
M --> C["AI Copilot"]
M --> ST["Settings"]

B --> F["FastAPI"]
B --> API["REST API"]
B --> WS["WebSocket"]
B --> DB["PostgreSQL"]

SS --> W["Wazuh"]
SS --> G["GitHub"]
SS --> SEM["Semgrep"]

W --> EV["Security Events"]
G --> EV
SEM --> EV

EV --> P
P --> FIND["Findings"]
P --> ALERT["Alerts"]
P --> SCAN["Scans"]
P --> RISK["Risk & Posture"]

F --> API
F --> WS
F --> DB

WS --> A
WS --> D

API --> D
API --> R
API --> A
API --> C

AI --> C
AI --> LLM["Cloud LLM"]
AI --> OFF["Offline Security Analysis"]

```

I --> REN["Render Cloud"]
I --> PG["Managed PostgreSQL"]
I --> HTTPS["HTTPS / WSS"]

SEC --> JWT["JWT Authentication"]
SEC --> RBAC["RBAC Authorization"]
SEC --> SECRET["Secret Management"]
SEC --> WEB["Webhook Validation"]
SEC --> AUDIT["Audit Logging"]

REN --> F
PG --> DB
HTTPS --> WS

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classDef mobile fill:#111827,stroke:#3b82f6,color:#ffffff;
classDef backend fill:#111827,stroke:#22c55e,color:#ffffff;
classDef security fill:#111827,stroke:#ef4444,color:#ffffff;
classDef ai fill:#111827,stroke:#a855f7,color:#ffffff;

class S main;
class M,D,R,A,C,ST mobile;
class B,F,API,WS,DB backend;
class SEC,JWT,RBAC,SECRET,WEB,AUDIT security;
class AI,LLM,OFF ai;

```

Comprehensive 17-Phase Product Roadmap

1. Vision & Requirements

- **Objective:** Define problem domain, user personas, operational constraints, and multi-tier cybersecurity monitoring requirements.
- **Tasks:**
 - Establish user stories for Tier-1/2/3 SOC Analysts, Incident Responders, and Security Leadership.
 - Define core latency benchmarks (< 500ms API response, < 50ms real-time event streaming).
 - Formulate air-gapped zero-latency Demo Mode specification for executive presentations.
- **Technology:** Markdown Specifications, Agile User Story Mapping.
- **Deliverables:** Project Charter, Functional Requirements Matrix, System Specification Dossier.
- **Validation:** Stakeholder review against real-world SOC operational friction points.
- **Status:**  COMPLETED

2. Architecture Design

- **Objective:** Formulate the multi-tier Clean Architecture, reactive state management model, and containerized cloud deployment topology.
 - **Tasks:**
 - Design 4-layer Feature-First Clean Architecture (Presentation, State, Domain, Data).
 - Establish Riverpod 2.5.1 provider dependencies and GoRouter 5-tab shell navigation.
 - Design dual-mode repository dispatching architecture (Demo Mode vs. Live Network Mode).
 - **Technology:** Flutter/Dart, Mermaid Diagrams, Clean Architecture Design Patterns.
 - **Deliverables:** docs/MINDMAP.md , Architecture Specification in docs/PROJECT_REPORT.md .
 - **Validation:** Architectural design review verifying decoupling of UI widgets from data layers.
 - **Status:**  COMPLETED
-

3. FastAPI Backend

- **Objective:** Develop an asynchronous, high-performance REST and WebSocket API microservice.
 - **Tasks:**
 - Implement route handlers (/health , /v1/auth , /v1/dashboard , /v1/repositories , /v1/soc/alerts , /v1/ai , /v1/reports).
 - Enforce Pydantic schema validation for all request and response envelopes.
 - Configure CORS middleware, timeout gates, and ASGI worker concurrency.
 - **Technology:** Python 3.11, FastAPI, Uvicorn, Pydantic, CORS Middleware.
 - **Deliverables:** Deployed FastAPI microservice on Render Cloud (<https://secureguard-backend-7eqm.onrender.com>).
 - **Validation:** Automated GET /health health probes and REST endpoint status verification.
 - **Status:**  COMPLETED
-

4. Database Persistence

- **Objective:** Implement cloud relational storage and client-side secure offline caching.
 - **Tasks:**
 - Provision managed PostgreSQL database with SSL connection pooling on cloud infrastructure.
 - Implement local key-value offline storage box (`HiveStorageService` , `securepulse_cache`).
 - Configure local cache synchronization for scan findings and executive posture scores.
 - **Technology:** PostgreSQL 16 (Cloud SSL), Hive 2.2.3 (Mobile Client).
 - **Deliverables:** `lib/core/storage/hive_storage_service.dart` , Cloud PostgreSQL database schemas.
 - **Validation:** Cold boot offline data recovery test validating instant cache hydration.
 - **Status:**  COMPLETED
-

5. Authentication & Credential Security

- **Objective:** Implement hardware-backed biometric authentication and stateless JWT session management.
 - **Tasks:**
 - Implement `local_auth` biometric challenge gate (Face ID / Android BiometricPrompt).
 - Integrate `FlutterSecureStorage` utilizing Android Keystore (AES-256 / RSA) and Apple Keychain.
 - Build `AuthRepositoryImpl` with login, logout, and token auto-injection interceptors.
 - **Technology:** `local_auth` 2.3.0 , `flutter_secure_storage` 9.2.4 , JWT (RFC 7519), bcrypt.
 - **Deliverables:** `lib/features/auth/` , `lib/core/services/biometric_service.dart` , `lib/core/storage/secure_storage_service.dart` .
 - **Validation:** Keystore read/write verification and biometric session recovery tests (`TC-06`).
 - **Status:**  COMPLETED
-

6. GitHub Integration

- **Objective:** Synchronize enterprise codebase telemetry, branch metadata, and security audit status.
- **Tasks:**
 - Implement `RepositoryModel` tracking language, branch, health grades (A–F), and vulnerability counts.
 - Build UI for monitored repositories list and repository detail views.
 - Integrate interactive in-app OAuth2 GitHub App consent deep-linking loop (`GithubOAuthService` , `securepulse://oauth/callback`).

- **Technology:** GitHub REST API v3, Webhooks, GoRouter, `app_links` .
 - **Deliverables:** `lib/features/repositories/` , `RepositoryRepositoryImpl` , `lib/core/services/github_oauth_service.dart` .
 - **Validation:** Repository inventory rendering, scan trigger dispatch (`TC-08`), and OAuth2 deep-link validation (`TC-16` – `TC-20`).
 - **Status:**  COMPLETED
-

7. Semgrep SAST Engine

- **Objective:** Ingest and categorize static application security testing findings and trigger on-demand scans.
 - **Tasks:**
 - Implement `ScanModel` and `FindingModel` with line numbers, file paths, and CWE tags.
 - Build scan detail screens with severity distribution badges (Critical, High, Medium, Low).
 - Wire single-tap "Trigger Scan" action dispatching `POST /v1/repositories/{id}/scan` .
 - **Technology:** Semgrep SAST JSON Schemas, Riverpod Providers.
 - **Deliverables:** `lib/features/scans/` , `lib/features/findings/` , `ScanRepository` , `FindingRepository` .
 - **Validation:** Verification of finding counts, CWE tags, and scan status state transitions.
 - **Status:**  COMPLETED
-

8. Wazuh SIEM Integration

- **Objective:** Ingest host intrusion detection logs, brute-force alarms, and system integrity events.
- **Tasks:**
 - Model SIEM alerts (`AlertModel`) capturing syslog payloads, attacker IP, and remediation advice.
 - Build Alert Detail screen with raw syslog inspection and "Quarantine IP" action button.
 - Implement bidirectional integration with Wazuh Manager API (`port 55000`) for cluster daemon monitoring (`wazuh-analysisd` , `wazuh-remoted` , `wazuh-modulesd`), agent inventory (`WazuhAgentModel`), and remote restart actions.
- **Technology:** Wazuh SIEM Syslog Parser, REST API, WebSockets, Riverpod State.
- **Deliverables:** `lib/features/alerts/` , `AlertsRepositoryImpl` , `WazuhRepositoryImpl` , `wazuh_models.dart` .

- **Validation:** Alert filtering by severity, agent inventory retrieval, daemon health probes, and remote restart dispatches (TC-09 , TC-23 – TC-27).
 - **Status:**  COMPLETED
-

9. Flutter Mobile Application

- **Objective:** Build a responsive, cyber-themed cross-platform mobile user interface.
 - **Tasks:**
 - Implement Material 3 Cyber Dark Obsidian (#0A0E1A) and Clean Light themes in `AppTheme` .
 - Build 5-tab shell navigation with animated navbar transitions (`SGNavbar`).
 - Implement executive dashboard with interactive posture gauge and `fl_chart` donut breakdown.
 - **Technology:** Flutter 3.x, Dart 3.x, GoRouter 13.2.0, `fl_chart` 0.66.2, Google Fonts.
 - **Deliverables:** Complete `lib/` codebase, `SecurePulseApp` root widget.
 - **Validation:** Widget pump testing (TC-15) and 60.0–120.0 fps frame rate profiling.
 - **Status:**  COMPLETED
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10. REST API Client & Interceptors

- **Objective:** Implement a resilient, typed HTTP network client with error classification.
 - **Tasks:**
 - Configure Dio 5.4.1 client with dynamic Bearer token headers and 12-second timeout gates.
 - Implement `ApiException` translation mapping 401, 403, 404, 500, and socket timeouts.
 - Add dynamic runtime `updateBaseUrl()` support for seamless environment switching.
 - **Technology:** Dio 5.4.1, Typed Dart Exceptions.
 - **Deliverables:** `lib/core/network/api_client.dart` , `lib/core/error/api_exception.dart` .
 - **Validation:** Unit tests verifying header injection, token removal, and base URL updates (TC-01 – TC-05).
 - **Status:**  COMPLETED
-

11. WebSocket Real-Time Engine

- **Objective:** Deliver sub-second SIEM threat streaming with automatic reconnection resilience.
 - **Tasks:**
 - Build `WebSocketService` with dynamic URL scheme conversion (`http->ws` , `https->wss`).
 - Broadcast incoming threat events to reactive Riverpod stream provider (`WebSocketAlertStreamProvider`).
 - Implement exponential backoff reconnect loops and HTTP long-polling fallback.
 - **Technology:** Dart `web_socket_channel 2.4.5` , StreamControllers, Riverpod Streams.
 - **Deliverables:** `lib/core/network/websocket_service.dart` .
 - **Validation:** Live scheme translation and Demo mode disconnection isolation tests (`TC-12` – `TC-14`).
 - **Status:**  COMPLETED
-

12. AI Security Copilot

- **Objective:** Provide conversational cybersecurity remediation advice and actionable code patches.
 - **Tasks:**
 - Build interactive streaming chat UI with markdown rendering and syntax-highlighted code cards.
 - Implement offline heuristic engine generating remediation patches for CVE-2024-3094, SQLi, and AWS secret leaks.
 - Connect live cloud LLM bridge via `POST /v1/ai/chat` for dynamic conversational analysis.
 - **Technology:** Flutter Markdown, Heuristic Rule Matchers, Cloud LLM Proxy.
 - **Deliverables:** `lib/features/ai/` , `AiRepositoryImpl` .
 - **Validation:** Heuristic code generation tests (`TC-10`) and interactive chat prompt rendering.
 - **Status:**  COMPLETED
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13. Security Hardening & Cryptographic Audit

- **Objective:** Protect client perimeter, vault credentials, and guarantee report non-repudiation.
- **Tasks:**
 - Implement STRIDE threat mitigations across all 6 attack categories.
 - Build `ReportPdfService` generating vector compliance PDFs (SOC 2, ISO 27001, PCI-DSS).
 - Embed on-device SHA256 cryptographic audit digest stamped onto all generated PDF pages.

- **Technology:** `pdf 3.12.0` , `crypto 3.0.3` (SHA256), Android Keystore, iOS Keychain.
 - **Deliverables:** `lib/core/services/report_pdf_service.dart` , Security Design in `docs/PROJECT_REPORT.md` .
 - **Validation:** Cryptographic audit digest verification and zero-leakage logging audit.
 - **Status:**  COMPLETED
-

14. Cloud Deployment & Containerization

- **Objective:** Automate cloud microservice deployment and provide containerized web distribution.
 - **Tasks:**
 - Create multi-stage `Dockerfile` compiling Flutter Web and serving via `nginx:alpine` on port 80.
 - Configure Render Cloud Web Service with Git auto-deploy webhooks and SSL/TLS termination.
 - Implement automated `/health` probes ensuring 99.9% service uptime.
 - **Technology:** Docker, Nginx, Render Cloud CI/CD, Let's Encrypt TLS 1.3.
 - **Deliverables:** `Dockerfile` , Render Web Service (`secureguard-backend-7eqm.onrender.com`).
 - **Validation:** Cloud health probe curl verification and Docker web container execution.
 - **Status:**  COMPLETED
-

15. Automated Testing & Verification

- **Objective:** Ensure 100% test pass rate across core business logic, API contracts, and widget trees.
 - **Tasks:**
 - Develop comprehensive test suite in `test/api_integration_test.dart` and `test/widget_test.dart` .
 - Execute static analysis audit via `flutter analyze` with zero warnings.
 - Benchmark runtime performance (frame rates, memory consumption, latency).
 - **Technology:** `flutter_test` , `flutter_lints 3.0.2` .
 - **Deliverables:** 15/15 passing automated test suite, `flutter analyze` clean report.
 - **Validation:** **15 / 15 Tests Passed (100% Pass Rate)**, `flutter analyze` (0 issues).
 - **Status:**  COMPLETED
-

16. Google Play Store Release Preparation

- **Objective:** Package, sign, and prepare the production Android release artifact for commercial distribution.
 - **Tasks:**
 - Compile optimized standalone release APK (`securepulse-release.apk` , **63.6 MB**) via `flutter build apk --release` .
 - Configure Android Gradle `signingConfigs` with automated fallback to debug keys for CI/development.
 - Implement ProGuard / R8 code shrinking and obfuscation rules (`android/app/proguard-rules.pro`).
 - Provide production keystore template (`android/key.properties.example`) for App Bundle (`AAB`) signing.
 - Prepare Google Play Console Data Safety declaration and privacy policy documentation.
 - **Technology:** Android Gradle Plugin, ProGuard / R8, Keytool RSA-2048, Google Play Console.
 - **Deliverables:** `securepulse-release.apk` , `android/key.properties.example` , `android/app/proguard-rules.pro` .
 - **Validation:** Successful Gradle compilation with R8 shrinking, sideload installation, and biometric execution (`TC-28` , `TC-29`).
 - **Status:**  COMPLETED
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17. Post-Launch Evolution & Offline Mutation Queue

- **Objective:** Evolve SecurePulse into an autonomous mobile security operations ecosystem with zero-downtime offline action durability.
- **Tasks:**
 - Build `OfflineQueueService` and `OfflineMutation` model with persistent Hive cache queueing.
 - Implement automated connectivity event listener (`connectivity_plus`) for auto-flushing queued mitigation actions upon network recovery.
 - Enhance `AlertsRepository` with transparent offline mutation enqueueing on network failures.
 - Register `offlineQueueServiceProvider` and `pendingMutationsStreamProvider` in Riverpod state tree.
- **Technology:** Hive Key-Value Box, `connectivity_plus` , Riverpod, Dio HTTP client.
- **Deliverables:** `lib/core/services/offline_queue_service.dart` , updated `AlertsRepositoryImpl` .
- **Validation:** Offline mutation enqueueing, background queue flushing, JSON serialization, and retry limit drop (`TC-30` – `TC-33`).

• **Status:**  COMPLETED

Summary of Phase Status

Category	Total Phases	Completed ()	In Progress ()	Planned ()	Not Implemented ()
Foundation & Architecture (Phases 1–5)	5	5	0	0	0
Toolchain Integrations (Phases 6–8)	3	3	0	0	0
Mobile & Real-Time Engine (Phases 9–12)	4	4	0	0	0
Production & Release (Phases 13–17)	5	5	0	0	0
OVERALL TOTAL	**17**	**17 (100.0%)**	**0 (0.0%)**	**0 (0.0%)**	**0 (0.0%)**
